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Experience

FlyRank

Machine Learning Engineering Intern

Remote

July 2026 – Present

- Architecting a scalable predictive ranking pipeline utilizing DuckDB and scikit-learn to process multi-gigabyte, enterprise search datasets for content discoverability.
- Designing robust model evaluation frameworks with strict data leakage controls, culminating in the deployment of a reproducible, public-facing content recommendation system.

Projects

⚙️ TernixEngine: Hardware-Optimized Edge Inference (Active Research) | C++20, AVX2 SIMD, Python

- Accelerated inference throughput for 1.58-bit ternary Large Language Models (LLMs), achieving an 8.2x speedup over scalar baselines, by engineering a dependency-free C++20 engine utilizing AVX2 SIMD branchless instructions.
- Eliminated floating-point bottlenecks and CPU branch predictor flushes, processing massively parallelized tensor transformations natively within LO registers, by replacing standard FP16 multiplications with in-register bitwise unpacking and integer additions.
- Established a highly scalable, hardware-native execution ecosystem, successfully exposing the C++ backend to high-level scripts via PyBind11, by architecting a strict 32-byte aligned tensor allocator wrapped in a unified CMake build matrix.

⚙️ VoltaSplat: Differentiable CUDA Rasterizer | C++20, CUDA C++, PyTorch

- Engineered a custom, differentiable 3D Gaussian Splatting rasterizer natively integrated into PyTorch via C++ ATen bindings, achieving a sub-17ms forward pass latency for 1M spatial points.
- Accelerated volumetric scene rendering throughput to 476+ FPS by architecting 16x16 tile-based rasterization kernels and implementing ultra-fast 64-bit depth sorting utilizing NVIDIA CUB primitives.
- Restricted peak VRAM utilization to strictly under 951MB during dense training passes by implementing thread-block '__shared__' memory caching and atomic gradient accumulation.

⚙️ nanoDist: Distributed ML Engine | NumPy, Distributed Systems, MLOps, FastAPI

- Slashed peak distributed training memory footprints by 68.07% (106MB to 34MB per node) by architecting a custom NumPy execution stack integrating ZeRO Stage 2 optimizer partitioning and Activation Checkpointing.
- Bounded inter-node communication overhead for scalable model training by deriving manual automatic differentiation and simulating peer-to-peer Ring All-Reduce primitives without external ML frameworks.
- Achieved production-grade MLOps integration by containerizing the distributed engine with Docker, orchestrating strict CI/CD validation pipelines, and exposing model inference via a serialized FastAPI REST server.

⚙️ HiveTorch: Federated Core | PyTorch, Federated Learning, Kubernetes, gRPC, MLOps

- Engineered a decentralized Federated Learning (FedAvg) engine, achieving a peak 87.0% global model accuracy across simulated edge environments, by deploying decoupled PyTorch microservices on Kubernetes via gRPC.
- Validated global convergence stability and algorithmic robustness under extreme data heterogeneity by implementing Dirichlet distribution sharding to mathematically synthesize complex, non-IID data topologies.
- Automated the end-to-end MLOps lifecycle, achieving fully reproducible experimental tracking and real-time operational telemetry, by integrating Weights & Biases, Prometheus, and Optuna into a Dockerized pipeline.

Technical Skills

Languages: C, C++ (C++17/20), CUDA C++, Triton, Python, Go, SQL

ML & AI Frameworks: PyTorch, JAX, ONNX, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV, Quantization (1.58-bit / GGUF)

Low-Level & Systems: GPU Programming, Hardware Intrinsics (SIMD/AVX2), Memory Management

Data, MLOps & Infrastructure: Kafka, Apache Spark, Kubernetes (K8s), Docker, MLflow, Optuna, AWS SageMaker, Distributed Data Parallel (DDP), CMake, Linux

Backend: gRPC, Redis, PostgreSQL, Elasticsearch, Node.js, Microservices

Education

Babu Banarasi Das University

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Lucknow, UP

Expected August 2027

- Relevant Coursework:** Data Science, Machine Learning, Linear Algebra, Calculus, Statistics, Probability, Data Structures, Design and Analysis of Algorithms, Database Management Systems, Computer Networking.

Achievements

- Secured **2nd Place at Masai x NoBroker HackArena 2025** for **CivicAgent**. Engineered an automated civic reporting platform featuring deterministic SLA tracking, transparent public dashboards, and scheduled escalation workflows to streamline municipal issue resolution.

Social Engagements & Leadership

Campus Partner: Perplexity AI (Fall '25 Cohort, Aug 2025 – Jan 2026)

Member: Google Developer Groups (GDG) Lucknow (Aug 2025 – Present)